MA 104: ODE – Tutorial 4 Solutions

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Problem 1

Consider the damped oscillator, driven by an external force $F_{ext}(t) = F_0(1 - e^{-\beta t})$ so that the ODE is

$$my'' + cy' + ky = F_0(1 - e^{-\beta t}),$$

where $\beta > 0$. Find the solution for the underdamped case assuming $y(0) = 0$ and $y'(0) = 0$. Analyze your solution for large β and very small β.

Solution

First, we solve the associated homogeneous equation describing the unforced oscillator:

$$my'' + cy' + ky = 0.$$

The characteristic equation is:

$$mr^2 + cr + k = 0.$$

The roots are found using the quadratic formula:

$$r = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}.$$

Since the problem specifies the **underdamped case**, we have $c^2 < 4mk$, which implies the discriminant is negative. We can write the roots as complex conjugates:

$$r = -\frac{c}{2m} \pm i\sqrt{\frac{k}{m} - \frac{c^2}{4m^2}}.$$

Let us define the damping coefficient γ and the damped natural frequency ω_d:

$$\gamma = \frac{c}{2m},$$
$$\omega_d = \sqrt{\frac{k}{m} - \gamma^2}.$$

Thus, the homogeneous solution is:

$$y_h(t) = e^{-\gamma t}(A\cos\omega_d t + B\sin\omega_d t).$$

We now look for a particular solution $y_p(t)$ corresponding to the non-homogeneous term

$$F_{ext}(t) = F_0 - F_0 e^{-\beta t}.$$

We can solve for F_0 and $-F_0e^{-\beta t}$ separately. For the constant force F_0, let $y_{p1} = K_1$ be the guess, for some constant K_1. Substituting into the ODE ($y'' = 0, y' = 0$):

$$\begin{aligned} m(0) + c(0) + k(K_1) &= F_0 \\ K_1 &= \frac{F_0}{k}. \end{aligned}$$

And for the exponential force $-F_0e^{-\beta t}$, let $y_{p2} = K_2e^{-\beta t}$ be the guess, for some constant K_2. We calculate the derivatives:

$$\begin{aligned} y'_{p2} &= -\beta K_2 e^{-\beta t}, \\ y''_{p2} &= \beta^2 K_2 e^{-\beta t}. \end{aligned}$$

Substituting these into the equation $my'' + cy' + ky = -F_0e^{-\beta t}$:

$$m(\beta^2 K_2 e^{-\beta t}) + c(-\beta K_2 e^{-\beta t}) + k(K_2 e^{-\beta t}) = -F_0 e^{-\beta t}.$$

Dividing both sides by $e^{-\beta t}$ (which is never zero):

$$K_2(m\beta^2 - c\beta + k) = -F_0.$$

Solving for K_2:

$$K_2 = \frac{-F_0}{m\beta^2 - c\beta + k}.$$

Thus the particular solution is $y_{p1} + y_{p2}$**:**

$$y_p(t) = \frac{F_0}{k} - \frac{F_0}{m\beta^2 - c\beta + k} e^{-\beta t},$$

and the general solution is $y(t) = y_h(t) + y_p(t)$:

$$y(t) = e^{-\gamma t}(A \cos \omega_d t + B \sin \omega_d t) + \frac{F_0}{k} - \frac{F_0}{m\beta^2 - c\beta + k} e^{-\beta t}.$$

Applying $y(0) = 0$:

$$\begin{aligned} y(0) = e^0(A \cdot 1 + B \cdot 0) + \frac{F_0}{k} - \frac{F_0}{m\beta^2 - c\beta + k} &= 0 \\ \implies A = F_0 \left(\frac{1}{m\beta^2 - c\beta + k} - \frac{1}{k} \right). \end{aligned}$$

Applying $y'(0) = 0$. First, differentiate $y(t)$:

$$y'(t) = -\gamma e^{-\gamma t}(A \cos \omega_d t + B \sin \omega_d t) + e^{-\gamma t}(-\omega_d A \sin \omega_d t + \omega_d B \cos \omega_d t) + \frac{\beta F_0}{m\beta^2 - c\beta + k} e^{-\beta t}.$$

At $t = 0$:

$$\begin{aligned} y'(0) = -\gamma A + \omega_d B + \frac{\beta F_0}{m\beta^2 - c\beta + k} &= 0 \\ \implies B &= \frac{1}{\omega_d} \left(\gamma F_0 \left(\frac{1}{m\beta^2 - c\beta + k} - \frac{1}{k} \right) - \frac{\beta F_0}{m\beta^2 - c\beta + k} \right) \\ &= \frac{F_0}{\omega_d} \left(-\frac{\gamma}{k} + \frac{\gamma - \beta}{m\beta^2 - c\beta + k} \right). \end{aligned}$$

Case 1: Large β (Fast Turn-On) When β is very large ($\beta \to \infty$), the external force term $F_0(1-e^{-\beta t})$ approaches a step function F_0 almost instantly. Mathematically, the term $\frac{1}{m\beta^2-c\beta+k}$ approaches zero. Consequently, the coefficients simplify:

$$A \approx -\frac{F_0}{k}, \quad B \approx -\frac{\gamma F_0}{k\omega_d}.$$

Substituting these into the general solution, the motion describes an underdamped oscillation about the new equilibrium position F_0/k:

$$y(t) \approx \frac{F_0}{k}\left[1 - e^{-\gamma t}\left(\cos\omega_d t + \frac{\gamma}{\omega_d}\sin\omega_d t\right)\right].$$

Physically, the force is applied so rapidly that the mass cannot respond instantly, causing it to overshoot and oscillate (ring) before settling at $y = F_0/k$.
Case 2: Small β (Slow Turn-On) When β is very small ($\beta \to 0$), the force increases gradually. We approximate $m\beta^2 - c\beta + k \approx k$. This causes the transient coefficients A and B to vanish:

$$A \approx F_0\left(\frac{1}{k} - \frac{1}{k}\right) = 0, \quad B \approx 0.$$

With the transient terms negligible, the solution is dominated by the particular integral:

$$y(t) \approx \frac{F_0}{k}(1 - e^{-\beta t}) = \frac{F_{ext}(t)}{k}.$$

Physically, this represents *quasi-static motion.* The system moves slowly enough that the inertia (my'') and damping (cy') forces are negligible; the spring force ky simply balances the external force at every instant, resulting in no oscillation.

Problem 2

(Higher order Cauchy-Euler equations) Let $D = \frac{d}{dx}$ and $\mathcal{D} = \frac{d}{dt}$ where $x = e^t$. Show

$$xD = \mathcal{D},$$

$$x^2D^2 = \mathcal{D}(\mathcal{D} - 1),$$

$$x^3D^3 = \mathcal{D}(\mathcal{D} - 1)(\mathcal{D} - 2).$$

Hence transform $(x^3D^3 + ax^2D^2 + bxD + c)y = 0$.

Solution

We are given the substitution $x = e^t$, which implies $t = \ln x$. The derivative of t with respect to x is:

$$\frac{dt}{dx} = \frac{1}{x}.$$

We use the Chain Rule to relate $D = \frac{d}{dx}$ and $\mathcal{D} = \frac{d}{dt}$:

$$Dy = \frac{dy}{dx} = \frac{dy}{dt}\frac{dt}{dx} = \frac{dy}{dt}\left(\frac{1}{x}\right) = \frac{1}{x}\mathcal{D}y.$$

Multiplying both sides by x, we obtain:

$$xDy = \mathcal{D}y \implies \boxed{xD = \mathcal{D}}.$$

We calculate $D^2y = \frac{d}{dx}\left(\frac{dy}{dx}\right)$ by differentiating the expression $\frac{1}{x}\mathcal{D}y$ with respect to x

$$\begin{aligned} D^2y &= \frac{d}{dx}\left(\frac{1}{x}\mathcal{D}y\right) \\ &= \frac{d}{dx}\left(\frac{1}{x}\right)\cdot \mathcal{D}y + \frac{1}{x}\cdot\frac{d}{dx}(\mathcal{D}y). \end{aligned}$$

Using $\frac{d}{dx} = \frac{1}{x}\frac{d}{dt}$:

$$\begin{aligned} D^2y &= \left(-\frac{1}{x^2}\right)\mathcal{D}y + \frac{1}{x}\left(\frac{1}{x}\frac{d}{dt}(\mathcal{D}y)\right) \\ &= -\frac{1}{x^2}\mathcal{D}y + \frac{1}{x^2}\mathcal{D}^2y \\ &= \frac{1}{x^2}(\mathcal{D}^2 - \mathcal{D})y. \end{aligned}$$

Multiplying by x^2:

$$x^2D^2 = \mathcal{D}(\mathcal{D}-1).$$

We differentiate $D^2y = \frac{1}{x^2}(\mathcal{D}^2 - \mathcal{D})y$ with respect to x:

$$\begin{aligned} D^3y &= \frac{d}{dx}\left[x^{-2}(\mathcal{D}^2-\mathcal{D})y\right] \\ &= \frac{d(x^{-2})}{dx}(\mathcal{D}^2-\mathcal{D})y + x^{-2}\frac{d}{dx}[(\mathcal{D}^2-\mathcal{D})y] \\ &= -2x^{-3}(\mathcal{D}^2-\mathcal{D})y + x^{-2}\left[\frac{1}{x}\frac{d}{dt}(\mathcal{D}^2-\mathcal{D})y\right] \\ &= -2x^{-3}(\mathcal{D}^2-\mathcal{D})y + x^{-3}(\mathcal{D}^3-\mathcal{D}^2)y. \end{aligned}$$

Factor out x^{-3}:

$$\begin{aligned} D^3y &= \frac{1}{x^3}\left[\mathcal{D}^3 - \mathcal{D}^2 - 2\mathcal{D}^2 + 2\mathcal{D}\right]y \\ &= \frac{1}{x^3}\left[\mathcal{D}^3 - 3\mathcal{D}^2 + 2\mathcal{D}\right]y. \end{aligned}$$

Factoring the operator polynomial:

$$\mathcal{D}^3 - 3\mathcal{D}^2 + 2\mathcal{D} = \mathcal{D}(\mathcal{D}^2 - 3\mathcal{D} + 2) = \mathcal{D}(\mathcal{D}-1)(\mathcal{D}-2).$$

Thus, multiplying by x^3:

$$x^3D^3 = \mathcal{D}(\mathcal{D}-1)(\mathcal{D}-2).$$

Substituting these into the given ODE $(x^3D^3 + ax^2D^2 + bxD + c)y = 0$:

$$[\mathcal{D}(\mathcal{D}-1)(\mathcal{D}-2) + a\mathcal{D}(\mathcal{D}-1) + b\mathcal{D} + c]y = 0.$$

Problem 3

Consider the n-th order linear equation

$$y^{(n)} + \sum_{i=0}^{n-1} a_i(x)y^{(i)} = r(x).$$

Assume that $y_1, ..., y_n$ are linearly independent solutions of the associated homogeneous equation. Prove that a particular integral is $y_p = \sum_{i=1}^{n} v_iy_i$, where $v_i' = \frac{R_i}{W}$.

Solution

Let the particular solution be $y_p(x) = \sum_{i=1}^{n} v_i(x)y_i(x)$. We need to determine the n unknown functions $v_i(x)$. To simplify the differentiation, we will impose $n-1$ constraints on the derivatives of v_i.

$$y_p' = \sum_{i=1}^{n}(v_i'y_i + v_iy_i').$$

Constraint 1: We assume $\sum_{i=1}^{n} v_i'y_i = 0$. So,

$$y_p' = \sum_{i=1}^{n} v_iy_i'.$$

The second derivative will then be

$$y_p'' = \sum_{i=1}^{n}(v_i'y_i' + v_iy_i'').$$

Constraint 2: We assume $\sum_{i=1}^{n} v_i'y_i' = 0$. So,

$$y_p'' = \sum_{i=1}^{n} v_iy_i''.$$

Continuing this process, for derivatives up to $n-1$, we impose:

$$\sum_{i=1}^{n} v_i'y_i^{(k)} = 0 \quad \text{for } k = 0, 1, \ldots, n-2.$$

This yields the simplified derivative forms:

$$y_p^{(k)} = \sum_{i=1}^{n} v_iy_i^{(k)} \quad \text{for } k = 0, \ldots, n-1.$$

For the final derivative $y_p^{(n)}$, we do not impose a constraint, so we differentiate the expression for $y_p^{(n-1)}$ normally:

$$y_p^{(n)} = \sum_{i=1}^{n} v_iy_i^{(n)} + \sum_{i=1}^{n} v_i'y_i^{(n-1)}.$$

Substituting y_p and its derivatives into the original ODE $y^{(n)} + \sum_{k=0}^{n-1} a_k(x)y^{(k)} = r(x)$:

$$\left(\sum_{i=1}^{n} v_iy_i^{(n)} + \sum_{i=1}^{n} v_i'y_i^{(n-1)}\right) + \sum_{k=0}^{n-1} a_k(x)\left(\sum_{i=1}^{n} v_iy_i^{(k)}\right) = r(x).$$

Rearranging the terms to group by v_i:

$$\sum_{i=1}^{n} v_i \underbrace{\left(y_i^{(n)} + \sum_{k=0}^{n-1} a_k(x)y_i^{(k)}\right)}_{\text{Zero, as } y_i \text{ are homogeneous solutions}} + \sum_{i=1}^{n} v_i'y_i^{(n-1)} = r(x).$$

Since each y_i is a solution to the homogeneous equation, the term in the parenthesis is zero. This leaves us with the final condition:

$$\sum_{i=1}^{n} v_i'y_i^{(n-1)} = r(x).$$

Combining our constraints and the final equation, we have a linear system for the unknowns $v_1', \dots, v_n'$:

$$\begin{aligned} v_1'y_1 + \cdots + v_n'y_n &= 0 \\ v_1'y_1' + \cdots + v_n'y_n' &= 0 \\ &\vdots \\ v_1'y_1^{(n-2)} + \cdots + v_n'y_n^{(n-2)} &= 0 \\ v_1'y_1^{(n-1)} + \cdots + v_n'y_n^{(n-1)} &= r(x). \end{aligned}$$

The coefficient matrix of this system is the Wronskian matrix W. Using Cramer's Rule, the solution for each v_i' is:

$$v_i' = \frac{\det(W_i)}{\det(W)} = \frac{R_i}{W},$$

where R_i (often denoted W_i) is the determinant of W with the i-th column replaced by the column vector $[0, 0, \dots, r(x)]^T$.

Remark: Illustration for $n = 3$ For a third-order equation ($n = 3$), the system for unknowns v_1', v_2', v_3' is:

$$\begin{pmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{pmatrix} \begin{pmatrix} v_1' \\ v_2' \\ v_3' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ r(x) \end{pmatrix}.$$

Here, the Wronskian W is the determinant of the coefficient matrix:

$$W = \det \begin{pmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{pmatrix}.$$

Then, the coefficients v_1, v_2, v_3 for the particular solution $y_p(x) = v_1y_1 + v_2y_2 + v_3y_3$ are obtained by integrating the expressions derived from Cramer's Rule:

$$v_1(x) = \int \frac{R_1}{W}\,dx, \quad v_2(x) = \int \frac{R_2}{W}\,dx, \quad v_3(x) = \int \frac{R_3}{W}\,dx,$$

where the numerators R_i are the determinants formed by replacing the i-th column of W with the vector $[0, 0, r(x)]^T$:

$$R_1 = \det \begin{pmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ r(x) & y_2'' & y_3'' \end{pmatrix}, \quad R_2 = \det \begin{pmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & r(x) & y_3'' \end{pmatrix}, \quad R_3 = \det \begin{pmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & r(x) \end{pmatrix}.$$

Problem 4

Solve the following non-homogeneous linear ODEs.

(a) $y'' - 4y' + 5y = e^{2x}\csc x$

Characteristic equation:

$$\begin{aligned} r^2 - 4r + 5 = 0 \\ r = \frac{4 \pm \sqrt{16 - 20}}{2} = 2 \pm i. \end{aligned}$$

Basis: $y_1 = e^{2x}\cos x$, $y_2 = e^{2x}\sin x$.
General solution: $y_h = e^{2x}(c_1\cos x + c_2\sin x)$. We calculate the Wronskian $W(y_1, y_2)$:

$$\begin{aligned} W &= \begin{vmatrix} e^{2x}\cos x & e^{2x}\sin x \\ (2e^{2x}\cos x - e^{2x}\sin x) & (2e^{2x}\sin x + e^{2x}\cos x) \end{vmatrix} \\ &= e^{4x}[\cos x(2\sin x + \cos x) - \sin x(2\cos x - \sin x)] \\ &= e^{4x}[2\sin x\cos x + \cos^2 x - 2\sin x\cos x + \sin^2 x] \\ &= e^{4x}(\cos^2 x + \sin^2 x) = e^{4x}. \end{aligned}$$

Now we find v_1' and v_2' using $r(x) = e^{2x}\csc x$:

$$\begin{aligned} v_1' &= \frac{-y_2 r(x)}{W} = \frac{-(e^{2x}\sin x)(e^{2x}\csc x)}{e^{4x}} = \frac{-e^{4x}(\sin x \cdot \frac{1}{\sin x})}{e^{4x}} = -1. \\ v_1 &= \int -1\, dx = -x. \\ v_2' &= \frac{y_1 r(x)}{W} = \frac{(e^{2x}\cos x)(e^{2x}\csc x)}{e^{4x}} = \frac{e^{4x}(\cos x \cdot \frac{1}{\sin x})}{e^{4x}} = \cot x. \\ v_2 &= \int \cot x\, dx = \ln|\sin x|. \end{aligned}$$

Hence

$$\begin{aligned} y(x) &= y_h + v_1 y_1 + v_2 y_2 \\ &= e^{2x}(c_1\cos x + c_2\sin x) - xe^{2x}\cos x + e^{2x}\sin x\ln|\sin x|. \end{aligned}$$

(b) $y'' + 6y' + 9y = \frac{16e^{-3x}}{x^2+1}$

Characteristic equation: $r^2 + 6r + 9 = (r+3)^2 = 0$.
Roots: $r = -3, -3$.
Basis: $y_1 = e^{-3x}$, $y_2 = xe^{-3x}$. $y_h = c_1 e^{-3x} + c_2 xe^{-3x}$. Calculate Wronskian W:

$$\begin{aligned} W &= \begin{vmatrix} e^{-3x} & xe^{-3x} \\ -3e^{-3x} & e^{-3x} - 3xe^{-3x} \end{vmatrix} \\ &= e^{-3x}(e^{-3x} - 3xe^{-3x}) - xe^{-3x}(-3e^{-3x}) \\ &= e^{-6x} - 3xe^{-6x} + 3xe^{-6x} = e^{-6x}. \end{aligned}$$

Calculate v_1, v_2:

$$\begin{aligned} v_1' &= \frac{-y_2 r(x)}{W} = \frac{-(xe^{-3x})\left(\frac{16e^{-3x}}{x^2+1}\right)}{e^{-6x}} = \frac{-16xe^{-6x}}{e^{-6x}(x^2+1)} = \frac{-16x}{x^2+1}. \\ v_1 &= -8\int \frac{2x}{x^2+1}dx = -8\ln(x^2+1). \\ v_2' &= \frac{y_1 r(x)}{W} = \frac{(e^{-3x})\left(\frac{16e^{-3x}}{x^2+1}\right)}{e^{-6x}} = \frac{16}{x^2+1}. \\ v_2 &= 16\int \frac{dx}{x^2+1} = 16\arctan x. \end{aligned}$$

General Solution:

$$y(x) = e^{-3x}(c_1 + c_2 x) - 8e^{-3x}\ln(x^2+1) + 16xe^{-3x}\arctan x.$$

(c) $y'' - y = \frac{1}{\cosh x}$

Characteristic equation: $r^2 - 1 = 0 \implies r = \pm 1$.
It is more convenient to use hyperbolic functions as basis. Basis: $y_1 = \cosh x$, $y_2 = \sinh x$.
Wronskian:

$$W = \begin{vmatrix} \cosh x & \sinh x \\ \sinh x & \cosh x \end{vmatrix} = \cosh^2 x - \sinh^2 x = 1.$$

Parameters:

$$\begin{aligned} v_1' &= \frac{-\sinh x \left(\frac{1}{\cosh x}\right)}{1} = -\tanh x. \\ v_1 &= \int -\frac{\sinh x}{\cosh x} dx = -\ln(\cosh x). \\ v_2' &= \frac{\cosh x \left(\frac{1}{\cosh x}\right)}{1} = 1. \\ v_2 &= \int 1\, dx = x. \end{aligned}$$

General Solution:

$$y(x) = c_1 \cosh x + c_2 \sinh x - \cosh x \ln(\cosh x) + x \sinh x.$$

Alternative Solution using Exponentials:
Basis: $y_1 = e^x$, $y_2 = e^{-x}$.
Wronskian:

$$W(e^x, e^{-x}) = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = (e^x)(-e^{-x}) - (e^{-x})(e^x) = -1 - 1 = -2.$$

Using Variation of Parameters with $r(x) = \frac{2}{e^x + e^{-x}}$:

$$\begin{aligned} v_1' &= \frac{-y_2 r(x)}{W} = \frac{-e^{-x}\left(\frac{2}{e^x+e^{-x}}\right)}{-2} = \frac{e^{-x}}{e^x + e^{-x}} = \frac{1}{e^{2x}+1}. \\ v_2' &= \frac{y_1 r(x)}{W} = \frac{e^{x}\left(\frac{2}{e^x+e^{-x}}\right)}{-2} = \frac{-e^{x}}{e^x + e^{-x}} = \frac{-e^{2x}}{e^{2x}+1}. \end{aligned}$$

Integrate v_1' (multiply numerator and denominator by e^{-2x}):

$$v_1 = \int \frac{e^{-2x}}{1 + e^{-2x}}\, dx = -\frac{1}{2}\ln(1 + e^{-2x}).$$

Integrate v_2' (let $u = e^{2x} + 1$):

$$v_2 = \int \frac{-e^{2x}}{e^{2x} + 1}\, dx = -\frac{1}{2}\ln(1 + e^{2x}).$$

Particular Solution:

$$\begin{aligned} y_p(x) &= v_1 y_1 + v_2 y_2 \\ &= -\frac{1}{2} e^x \ln(1 + e^{-2x}) - \frac{1}{2} e^{-x} \ln(1 + e^{2x}). \end{aligned}$$

General Solution:

$$y(x) = c_1 e^x + c_2 e^{-x} - \frac{1}{2}\left[e^x \ln(1 + e^{-2x}) + e^{-x}\ln(1 + e^{2x})\right].$$

Note: This is equivalent to the hyperbolic form $-x \sinh x + \cosh x \ln(\cosh x)$.

(d) $y''' - y'' + y' - y = e^{-t} \sin t$

Characteristic equation: $r^3 - r^2 + r - 1 = r^2(r-1) + 1(r-1) = (r^2+1)(r-1) = 0$.
Roots: $r = 1, r = i, r = -i$.
Basis: $e^t, \cos t, \sin t$.
RHS is $e^{-t} \sin t$. Guess: $y_p = e^{-t}(A \cos t + B \sin t)$.
We calculate derivatives:

$$\begin{aligned}
y_p' &= -e^{-t}(A\cos t + B\sin t) + e^{-t}(-A\sin t + B\cos t) \\
&= e^{-t}[(B-A)\cos t - (A+B)\sin t]. \\
y_p'' &= -e^{-t}[(B-A)\cos t - (A+B)\sin t] + e^{-t}[-(B-A)\sin t - (A+B)\cos t] \\
&= e^{-t}[(A-B-A-B)\cos t + (A+B-B+A)\sin t] \\
&= e^{-t}[-2B\cos t + 2A\sin t]. \\
y_p''' &= -e^{-t}[-2B\cos t + 2A\sin t] + e^{-t}[2B\sin t + 2A\cos t] \\
&= e^{-t}[(2B+2A)\cos t + (2B-2A)\sin t].
\end{aligned}$$

Substitute into $y''' - y'' + y' - y = e^{-t} \sin t$:

$$\begin{aligned}
\text{Coeff of } e^{-t}\cos t : &\quad (2A+2B) - (-2B) + (B-A) - A = 5B. \\
\text{Coeff of } e^{-t}\sin t : &\quad (2B-2A) - (2A) - (A+B) - B = -5A.
\end{aligned}$$

$5B = 0 \implies B = 0$. $-5A = 1 \implies A = -1/5$.
General Solution

$$y(t) = c_1 e^t + c_2 \cos t + c_3 \sin t - \frac{1}{5} e^{-t} \cos t.$$

(e) $y''' + y' = \sec t$

Characteristic equation: $r^3 + r = r(r^2+1) = 0 \implies r = 0, \pm i$.
Basis: $y_1 = 1, y_2 = \cos t, y_3 = \sin t$.
Wronskian W:

$$W = \begin{vmatrix} 1 & \cos t & \sin t \\ 0 & -\sin t & \cos t \\ 0 & -\cos t & -\sin t \end{vmatrix} = 1 \begin{vmatrix} -\sin t & \cos t \\ -\cos t & -\sin t \end{vmatrix} = (\sin^2 t + \cos^2 t) = 1.$$

The non-homogeneous term vector is $[0, 0, \sec t]^T$.

$$\begin{aligned}
v_1' &= \frac{1}{1} \begin{vmatrix} 0 & \cos t & \sin t \\ 0 & -\sin t & \cos t \\ \sec t & -\cos t & -\sin t \end{vmatrix} = \sec t(\cos^2 t + \sin^2 t) = \sec t. \\
v_1 &= \int \sec t \, dt = \ln|\sec t + \tan t|.
\end{aligned}$$

$$\begin{aligned}
v_2' &= \frac{1}{1} \begin{vmatrix} 1 & 0 & \sin t \\ 0 & 0 & \cos t \\ 0 & \sec t & -\sin t \end{vmatrix} = -\sec t(\cos t) = -1. \\
v_2 &= -t.
\end{aligned}$$

$$\begin{aligned}
v_3' &= \frac{1}{1} \begin{vmatrix} 1 & \cos t & 0 \\ 0 & -\sin t & 0 \\ 0 & -\cos t & \sec t \end{vmatrix} = \sec t(-\sin t) = -\tan t. \\
v_3 &= -\int \tan t \, dt = \ln|\cos t|.
\end{aligned}$$

General Solution:

$$y(t) = c_1 + c_2 \cos t + c_3 \sin t + \ln|\sec t + \tan t| - t\cos t + \sin t \ln|\cos t|.$$

Problem 5

Solve the IVP: $x^3y''' + xy' - y = x^2$**, with** $y(1) = 1$**,** $y'(1) = 3$**,** $y''(1) = 14$.

Solution

Using the results from Problem 2 ($x = e^t$):

$$\begin{aligned} x^3D^3 &\to \mathcal{D}(\mathcal{D}-1)(\mathcal{D}-2) \\ xD &\to \mathcal{D} \\ y &\to 1. \end{aligned}$$

The ODE becomes:

$$\begin{aligned} [\mathcal{D}(\mathcal{D}-1)(\mathcal{D}-2) + \mathcal{D} - 1]y &= (e^t)^2 \\ [\mathcal{D}(\mathcal{D}^2 - 3\mathcal{D} + 2) + \mathcal{D} - 1]y &= e^{2t} \\ [\mathcal{D}^3 - 3\mathcal{D}^2 + 3\mathcal{D} - 1]y &= e^{2t}. \end{aligned}$$

Recognizing the polynomial expansion:

$$(\mathcal{D}-1)^3 y = e^{2t}.$$

Homogeneous roots of $(r-1)^3 = 0$ are $r = 1, 1, 1$.

$$y_h(t) = c_1e^t + c_2te^t + c_3t^2e^t.$$

Particular solution for e^{2t}: Try $y_p = Ae^{2t}$.

$$(2-1)^3Ae^{2t} = e^{2t} \implies 1^3 \cdot A = 1 \implies A = 1.$$

Thus, $y(t) = (c_1 + c_2t + c_3t^2)e^t + e^{2t}$. Substitute $t = \ln x$ and $e^t = x$:

$$y(x) = x(c_1 + c_2 \ln x + c_3(\ln x)^2) + x^2.$$

1. $y(1) = 1$:

$$1(c_1 + 0 + 0) + 1 = 1 \implies c_1 + 1 = 1 \implies \boxed{c_1 = 0}.$$

   Now $y(x) = c_2 x \ln x + c_3 x(\ln x)^2 + x^2$.

2. $y'(1) = 3$:

$$\begin{aligned} y'(x) &= c_2(1 + \ln x) + c_3(2\ln x + (\ln x)^2 + 2\ln x) + 2x \quad \text{(Wait, differentiating properly)} \\ y(x) &= c_2x\ln x + c_3x(\ln x)^2 + x^2 \\ y'(x) &= c_2(\ln x + x \cdot \frac{1}{x}) + c_3((\ln x)^2 + x \cdot 2\ln x \cdot \frac{1}{x}) + 2x \\ &= c_2(\ln x + 1) + c_3((\ln x)^2 + 2\ln x) + 2x. \\ y'(1) &= c_2(0+1) + c_3(0+0) + 2 = 3 \implies c_2 + 2 = 3 \implies \boxed{c_2 = 1}. \end{aligned}$$

3. $y''(1) = 14$:

$$\begin{aligned} y''(x) &= c_2(\frac{1}{x}) + c_3(2\ln x \cdot \frac{1}{x} + \frac{2}{x}) + 2 \\ &= \frac{c_2}{x} + \frac{2c_3(\ln x + 1)}{x} + 2. \\ y''(1) &= \frac{1}{1} + \frac{2c_3(0+1)}{1} + 2 = 14 \\ &= 1 + 2c_3 + 2 = 14 \implies 2c_3 = 11 \implies \boxed{c_3 = 11/2}. \end{aligned}$$

Thus,

$$y(x) = x\ln x + \frac{11}{2}x(\ln x)^2 + x^2.$$

Problem 6

Determine intervals in which solutions are sure to exist.
(a) $y^{(4)} + 4y''' + 3y = t^2$
(b) $t\ y''' + (sin\ t)y'' + 3y = cos\ t$

Solution

The Existence and Uniqueness Theorem for linear ODEs states that for an equation of the form:

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \cdots + p_0(t)y = g(t)$$

a unique solution exists on any interval I where all coefficients $p_i(t)$ and $g(t)$ are continuous.

(a) The equation is already in standard form (leading coefficient is 1):

$$y^{(4)} + 4y''' + 0y'' + 0y' + 3y = t^2.$$

The coefficients are constant functions $4, 0, 0, 3$, which are continuous everywhere. The forcing function $g(t) = t^2$ is a polynomial, also continuous everywhere. Thus, the solution exists on the entire real line:

$$I = (-\infty, \infty).$$

(b) The leading coefficient is t. To apply the theorem, we put the equation in standard form:

$$y''' + \frac{\sin t}{t}y'' + 0y' + \frac{3}{t}y = \frac{\cos t}{t}.$$

We analyze the continuity of the terms:

- $p_2(t) = \frac{\sin t}{t}$. At $t = 0$, the limit is 1, so the discontinuity is removable, but formally the term is undefined at $t = 0$ in the algebraic sense.
- $p_0(t) = \frac{3}{t}$. Infinite discontinuity at $t = 0$.
- $g(t) = \frac{\cos t}{t}$. Infinite discontinuity at $t = 0$.

Since the coefficients and forcing function are discontinuous at $t = 0$, the existence of a solution is only guaranteed on intervals that do not contain $t = 0$.

$$I_1 = (-\infty, 0) \quad \text{or} \quad I_2 = (0, \infty).$$

Problem 7

Use the method of reduction of order ($y = u(x)y_1(x)$) to solve $(2-t)y''' + (2t-3)y'' - ty' + y = 0$, $t < 2$; $y_1(t) = e^t$.

Solution

Let $y(t) = u(t)e^t$. We compute the derivatives using the product rule:

$$\begin{aligned} y' &= e^t(u' + u) \\ y'' &= e^t(u'' + u' + u' + u) = e^t(u'' + 2u' + u) \\ y''' &= e^t(u''' + 2u'' + u'' + u' + 2u' + u) = e^t(u''' + 3u'' + 3u' + u). \end{aligned}$$

Substitute these into the ODE. Notice that every term has a factor of e^t, which is never zero, so we can divide it out immediately:

$$(2-t)(u''' + 3u'' + 3u' + u) + (2t-3)(u'' + 2u' + u) - t(u' + u) + u = 0.$$

Now we group terms by the derivatives of u:

$$\begin{aligned} \text{Coeff of } u''' &: (2-t) \\ \text{Coeff of } u'' &: 3(2-t) + (2t-3) = 6 - 3t + 2t - 3 = 3 - t \\ \text{Coeff of } u' &: 3(2-t) + 2(2t-3) - t = 6 - 3t + 4t - 6 - t = 0 \\ \text{Coeff of } u &: (2-t) + (2t-3) - t + 1 = 2 - t + 2t - 3 - t + 1 = 0. \end{aligned}$$

The u and u' terms vanish! The equation reduces to:

$$(2-t)u''' + (3-t)u'' = 0.$$

Let $w = u''$. This becomes a first-order separable ODE:

$$\begin{aligned} (2-t)w' + (3-t)w &= 0 \\ \frac{w'}{w} &= -\frac{3-t}{2-t} = \frac{t-3}{2-t} = \frac{-(2-t)-1}{2-t} = -1 - \frac{1}{2-t}. \end{aligned}$$

Integrating with respect to t:

$$\begin{aligned} \ln|w| &= \int \left(-1 - \frac{1}{2-t}\right) dt = -t + \int \frac{-dt}{2-t} \\ &= -t + \ln|2-t| + C_1. \end{aligned}$$

Exponentiating:

$$w = A(2-t)e^{-t}.$$

Since $w = u''$, we integrate twice to find u. First integration:

$$u' = \int A(2-t)e^{-t}dt = A\left[\int 2e^{-t}dt - \int te^{-t}dt\right].$$

Using $\int te^{-t}dt = -te^{-t} - e^{-t}$:

$$u' = A[-2e^{-t} - (-te^{-t} - e^{-t})] = A[-e^{-t} + te^{-t}] = Ae^{-t}(t-1) + B.$$

Second integration:

$$u = \int [A(t-1)e^{-t} + B]dt = A \int (t-1)e^{-t}dt + Bt + C.$$

Using integration by parts on $(t-1)e^{-t}$:

$$\int (t-1)e^{-t}dt = -(t-1)e^{-t} + \int e^{-t}dt = -(t-1)e^{-t} - e^{-t} = -te^{-t}.$$

Thus:

$$u = -Ate^{-t} + Bt + C.$$

The general solution is $y = ue^t$:

$$\begin{aligned} y(t) &= (-Ate^{-t} + Bt + C)e^t \\ &= -At + Bte^t + Ce^t. \end{aligned}$$

Renaming constants, the general solution is:

$$y(t) = c_1 t + c_2 te^t + c_3 e^t.$$

Problem 8

Determine whether the given functions are linearly dependent or linearly independent.

(a) $f_1(t) = 2t - 3, \quad f_2(t) = 2t^2 + 1, \quad f_3(t) = 3t^2 + t$.

To define linear dependence, we check if there exist constants c_1, c_2, c_3 not all zero such that:

$$c_1(2t-3) + c_2(2t^2+1) + c_3(3t^2+t) = 0 \quad \text{for all } t.$$

Grouping by powers of t:

$$(2c_2 + 3c_3)t^2 + (2c_1 + c_3)t + (-3c_1 + c_2) = 0.$$

This implies the system:

1. $2c_2 + 3c_3 = 0$
2. $2c_1 + c_3 = 0 \implies c_3 = -2c_1$
3. $-3c_1 + c_2 = 0 \implies c_2 = 3c_1$

Substitute (2) and (3) into (1):

$$2(3c_1) + 3(-2c_1) = 6c_1 - 6c_1 = 0.$$

The system is consistent for any value of c_1. Let $c_1 = 1$. Then $c_2 = 3$ and $c_3 = -2$. Check:

$$1(2t-3) + 3(2t^2+1) - 2(3t^2+t) = 2t - 3 + 6t^2 + 3 - 6t^2 - 2t = 0.$$

Since a non-trivial combination exists, the functions are **Linearly Dependent**.

(b) $f_1(t) = 2t - 3, \quad f_2(t) = t^2 + 1, \quad f_3(t) = 2t^2 - t, \quad f_4(t) = t^2 + t + 1$

These are four vectors in the vector space of polynomials of degree at most 2 (P_2). The dimension of P_2 is 3 (basis: $\{1, t, t^2\}$). Any set of n vectors in an m-dimensional space where $n > m$ must be linearly dependent. Since $4 > 3$, these functions are **Linearly Dependent**.

Problem 10

Given $W(y_1(2), y_2(2)) = \frac{1}{32}$ for $x^2y'' + 5xy' + 4y = x^2 sin\ x$, find $W(x)$.

Solution

Abel's Formula states that for a second-order linear ODE $y'' + P(x)y' + Q(x)y = 0$, the Wronskian $W(x)$ is given by:

$$W(x) = C \exp\left(-\int P(x)dx\right).$$

First, we must put the given ODE in standard form by dividing by the leading coefficient x^2:

$$y'' + \frac{5}{x}y' + \frac{4}{x^2}y = \sin x.$$

Identify $P(x) = \frac{5}{x}$. Now, compute the integral:

$$\int P(x)dx = \int \frac{5}{x}dx = 5 \ln x = \ln(x^5).$$

Substitute into Abel's formula:

$$W(x) = C \exp(-\ln(x^5)) = C \exp(\ln(x^{-5})) = Cx^{-5} = \frac{C}{x^5}.$$

We are given the initial condition $W(2) = \frac{1}{32}$. Substitute $x = 2$:

$$W(2) = \frac{C}{2^5} = \frac{C}{32}.$$

Equating the values:

$$\frac{C}{32} = \frac{1}{32} \implies C = 1.$$

Thus, the Wronskian is:

$$W(x) = \frac{1}{x^5}.$$

Problem 11

Consider the ODE $y'' + \frac{1-\alpha^2}{4x^2}y = 0$. Comment on the nature of the solution for $\alpha = \frac{1}{n}$.

Solution

Multiply the equation by x^2 to recognize it as a Cauchy-Euler equation:

$$x^2y'' + \frac{1-\alpha^2}{4}y = 0.$$

We assume a solution of the form $y = x^r$. Substituting into the ODE leads to the characteristic equation:

$$\begin{aligned} r(r-1) + \frac{1-\alpha^2}{4} &= 0 \\ r^2 - r + \frac{1}{4} - \frac{\alpha^2}{4} &= 0 \\ \left(r - \frac{1}{2}\right)^2 &= \frac{\alpha^2}{4}. \end{aligned}$$

Taking the square root:

$$r - \frac{1}{2} = \pm\frac{\alpha}{2} \implies r = \frac{1 \pm \alpha}{2}.$$

Substitute $\alpha = 1/n$:

$$r_1 = \frac{1 + 1/n}{2} = \frac{n+1}{2n},$$
$$r_2 = \frac{1 - 1/n}{2} = \frac{n-1}{2n}.$$

Since $\alpha \neq 0$, the roots are distinct and real. The general solution is:

$$y(x) = c_1 x^{\frac{n+1}{2n}} + c_2 x^{\frac{n-1}{2n}}.$$

Nature of Solution: The solutions are algebraic power functions. They are non-oscillatory (unlike sine/cosine solutions which appear when roots are complex).

Problem 12

Find the general solution of: (a) $y'' - y' - 2y = 4x^2$ **(b)** $y'' + 10y' + 25y = 14e^{-5x}$

Solution (a)

Characteristic equation: $r^2 - r - 2 = (r-2)(r+1) = 0$. Roots: $2, -1$.

$$y_h = c_1 e^{2x} + c_2 e^{-x}.$$

RHS is $4x^2$ (polynomial of degree 2). Guess $y_p = Ax^2 + Bx + C$.

$$y_p' = 2Ax + B,$$
$$y_p'' = 2A.$$

Substitute into ODE:

$$(2A) - (2Ax + B) - 2(Ax^2 + Bx + C) = 4x^2$$
$$-2Ax^2 + (-2A - 2B)x + (2A - B - 2C) = 4x^2.$$

Compare coefficients:

1. x^2: $-2A = 4 \implies A = -2$.
2. x^1: $-2(-2) - 2B = 0 \implies 4 - 2B = 0 \implies B = 2$.
3. x^0: $2(-2) - 2 - 2C = 0 \implies -6 - 2C = 0 \implies C = -3$.

$$y(x) = c_1 e^{2x} + c_2 e^{-x} - 2x^2 + 2x - 3.$$

Solution (b)

Characteristic equation: $r^2 + 10r + 25 = (r + 5)^2 = 0$. Double root $r = -5$.

$$y_h = c_1 e^{-5x} + c_2 x e^{-5x}.$$

RHS is $14e^{-5x}$. Normal guess would be Ae^{-5x}. But e^{-5x} is a root. Multiply by x: Axe^{-5x}. But xe^{-5x} is also a root (double root case). Multiply by x again: $y_p = Ax^2 e^{-5x}$. Derivatives:

$$\begin{aligned} y_p' &= A(2xe^{-5x} - 5x^2 e^{-5x}) = Ae^{-5x}(2x - 5x^2). \\ y_p'' &= Ae^{-5x}[(2 - 10x) - 5(2x - 5x^2)] = Ae^{-5x}(25x^2 - 20x + 2). \end{aligned}$$

Substitute into ODE:

$$\begin{aligned} Ae^{-5x}[(25x^2 - 20x + 2) + 10(2x - 5x^2) + 25x^2] &= 14e^{-5x} \\ A[25x^2 - 20x + 2 + 20x - 50x^2 + 25x^2] &= 14 \\ A[0x^2 + 0x + 2] &= 14 \\ 2A = 14 &\implies A = 7. \end{aligned}$$

$$y(x) = (c_1 + c_2 x + 7x^2)e^{-5x}.$$

Problem 13

(a) Show that applying variation of parameters to $y'' + y = f(x)$ leads to $y_p(x) = \int_0^x f(t)sin(x-t)dt$. (b) Similar formula for $y'' + k^2 y = f(x)$.

Solution (a)

Basis: $y_1 = \cos x$, $y_2 = \sin x$. Wronskian $W = \cos^2 x + \sin^2 x = 1$. Using the formula:

$$\begin{aligned} y_p(x) &= -y_1(x) \int_0^x \frac{y_2(t)f(t)}{W} dt + y_2(x) \int_0^x \frac{y_1(t)f(t)}{W} dt \\ &= -\cos x \int_0^x \sin t f(t) dt + \sin x \int_0^x \cos t f(t) dt \\ &= \int_0^x (\sin x \cos t - \cos x \sin t) f(t) dt. \end{aligned}$$

Using the identity $\sin(A - B) = \sin A \cos B - \cos A \sin B$:

$$y_p(x) = \int_0^x \sin(x - t) f(t) dt.$$

Solution (b)

For $y'' + k^2 y = f(x)$, roots are $\pm ki$. Basis: $y_1 = \cos kx, y_2 = \sin kx$. Wronskian:

$$W = k\cos^2 kx + k\sin^2 kx = k.$$

Formula:

$$\begin{aligned} y_p(x) &= \frac{-\cos kx}{k} \int_0^x \sin kt f(t) dt + \frac{\sin kx}{k} \int_0^x \cos kt f(t) dt \\ &= \frac{1}{k} \int_0^x (\sin kx \cos kt - \cos kx \sin kt) f(t) dt \\ &= \frac{1}{k} \int_0^x \sin(k(x - t)) f(t) dt. \end{aligned}$$

Problem 14

Find the general solution of: (a) $(1-x)y'' + xy' - y = (1-x)^2$ **(b)** $xy'' - (1+x)y' + y = x^2e^{2x}$ **(c)** $y'' + 4y = 2\ cos\ 2x + 10e^x$ **(d)** $y'' - y = e^{-x}(sin\ x + cos\ x)$ **(e)** $y''' - 3y'' - y' + 3y = x^2e^x$

Solution (a)

By inspection, $y_1 = x$ works $(0 + x(1) - x = 0)$.
By inspection, $y_2 = e^x$ works $((1-x)e^x + xe^x - e^x = e^x - xe^x + xe^x - e^x = 0)$.
Wronskian: $W = xe^x - 1e^x = e^x(x-1)$.
Standard form RHS: Divide by $(1-x)$.
$r(x) = \frac{(1-x)^2}{1-x} = 1 - x$.

$$\begin{aligned} v_1' &= \frac{-e^x(1-x)}{e^x(x-1)} = \frac{e^x(x-1)}{e^x(x-1)} = 1 \implies v_1 = x. \\ v_2' &= \frac{x(1-x)}{e^x(x-1)} = \frac{-x(x-1)}{e^x(x-1)} = -xe^{-x}. \end{aligned}$$

Integrate v_2' by parts $(u = x, dv = e^{-x}dx)$:

$$v_2 = -[-xe^{-x} - \int -e^{-x}dt] = xe^{-x} + e^{-x} = e^{-x}(x+1).$$

Particular Solution $v_1y_1 + v_2y_2$:

$$y_p = x(x) + e^x[e^{-x}(x+1)] = x^2 + x + 1.$$

General Solution:

$$y = c_1x + c_2e^x + x^2 + x + 1.$$

Solution (b)

By inspection, $y_1 = 1 + x$ $(x(0) - (1+x)(1) + (1+x) = 0)$.
By inspection, $y_2 = e^x$ $(xe^x - (1+x)e^x + e^x = 0)$.
Wronskian $W = (1+x)e^x - 1(e^x) = xe^x$.
Standard form RHS: Divide by x.
$r(x) = xe^{2x}$.

$$\begin{aligned} v_1' &= \frac{-e^x(xe^{2x})}{xe^x} = -e^{2x} \implies v_1 = -\frac{1}{2}e^{2x}. \\ v_2' &= \frac{(1+x)(xe^{2x})}{xe^x} = (1+x)e^x = e^x + xe^x. \end{aligned}$$

Integrate v_2':

$$v_2 = e^x + (xe^x - e^x) = xe^x.$$

Particular Solution $v_1y_1 + v_2y_2$:

$$\begin{aligned} y_p &= -\frac{1}{2}e^{2x}(1+x) + e^x(xe^x) = -\frac{1}{2}e^{2x} - \frac{1}{2}xe^{2x} + xe^{2x} \\ &= \frac{1}{2}xe^{2x} - \frac{1}{2}e^{2x} = \frac{1}{2}e^{2x}(x-1). \end{aligned}$$

General Solution:

$$y = c_1(1+x) + c_2e^x + \frac{1}{2}e^{2x}(x-1).$$

Solution (c)

Characteristic equation: $r^2 + 4 = 0 \implies r = \pm 2i$.
Complementary function: $y_c = c_1 \cos 2x + c_2 \sin 2x$.
RHS: $2\cos 2x$ (resonant) and $10e^x$ (non-resonant).
Guess form: $y_p = x(A\cos 2x + B\sin 2x) + Ce^x$.

$$\begin{aligned} y_p' &= A\cos 2x + B\sin 2x + 2x(-A\sin 2x + B\cos 2x) + Ce^x. \\ y_p'' &= -4A\sin 2x + 4B\cos 2x - 4x(A\cos 2x + B\sin 2x) + Ce^x. \end{aligned}$$

Substitute into $y'' + 4y$:

$$(-4A\sin 2x + 4B\cos 2x) + 5Ce^x = 2\cos 2x + 10e^x.$$

Equating coefficients:

$$\begin{aligned} 5C = 10 &\implies C = 2. \\ -4A = 0 &\implies A = 0. \\ 4B = 2 &\implies B = 1/2. \end{aligned}$$

General Solution:

$$y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{2} x \sin 2x + 2e^x.$$

Solution (d)

Characteristic equation: $r^2 - 1 = 0 \implies r = \pm 1$.
Complementary function: $y_c = c_1 e^x + c_2 e^{-x}$.
Guess form: $y_p = e^{-x}(A\cos x + B\sin x)$.

$$\begin{aligned} y_p' &= e^{-x}[(B - A)\cos x - (A + B)\sin x]. \\ y_p'' &= e^{-x}[-2B\cos x + 2A\sin x]. \end{aligned}$$

Substitute into $y'' - y$:

$$\begin{aligned} e^{-x}(-2B\cos x + 2A\sin x) - e^{-x}(A\cos x + B\sin x) &= e^{-x}(\sin x + \cos x) \\ e^{-x}[(-2B - A)\cos x + (2A - B)\sin x] &= e^{-x}(\cos x + \sin x). \end{aligned}$$

Equating coefficients:

$$\begin{aligned} -A - 2B &= 1 \\ 2A - B &= 1 \implies B = 2A - 1. \end{aligned}$$

Substitute B into first equation:

$$\begin{aligned} -A - 2(2A - 1) = 1 \implies -5A + 2 = 1 &\implies A = 1/5. \\ B = 2(1/5) - 1 &= -3/5. \end{aligned}$$

General Solution:

$$y = c_1 e^x + c_2 e^{-x} + \frac{1}{5} e^{-x}(\cos x - 3\sin x).$$

Solution (e)

Characteristic equation: $r^3 - 3r^2 - r + 3 = 0 \implies (r-1)(r+1)(r-3) = 0$.
Complementary function: $y_c = c_1e^x + c_2e^{-x} + c_3e^{3x}$.
RHS: x^2e^x. Since e^x is a solution ($r = 1$), multiply guess by x.
Guess: $y_p = x(Ax^2 + Bx + C)e^x = (Ax^3 + Bx^2 + Cx)e^x$.
Let $P(x) = Ax^3 + Bx^2 + Cx$. Then $y_p = Pe^x$.

$$\begin{aligned} y_p' &= (P' + P)e^x \\ y_p'' &= (P'' + 2P' + P)e^x \\ y_p''' &= (P''' + 3P'' + 3P' + P)e^x \end{aligned}$$

Substitute into $y''' - 3y'' - y' + 3y$:

$$\begin{aligned} e^x[(P''' + 3P'' + 3P' + P) - 3(P'' + 2P' + P) - (P' + P) + 3P] &= x^2e^x \\ e^x[P''' - 4P'] &= x^2e^x \\ P''' - 4P' &= x^2. \end{aligned}$$

Differentiating $P(x) = Ax^3 + Bx^2 + Cx$:

$$\begin{aligned} P' &= 3Ax^2 + 2Bx + C \\ P''' &= 6A. \end{aligned}$$

Substitute into $P''' - 4P' = x^2$:

$$\begin{aligned} 6A - 4(3Ax^2 + 2Bx + C) &= x^2 \\ -12Ax^2 - 8Bx + (6A - 4C) &= x^2. \end{aligned}$$

Equating coefficients:

$$\begin{aligned} -12A = 1 &\implies A = -1/12. \\ -8B = 0 &\implies B = 0. \\ 6A - 4C = 0 &\implies 4C = 6(-1/12) = -1/2 \implies C = -1/8. \end{aligned}$$

General Solution:

$$y = c_1e^x + c_2e^{-x} + c_3e^{3x} - \left(\frac{1}{12}x^3 + \frac{1}{8}x\right)e^x.$$

Problem 15

Show that $\phi(x) = \alpha y_1(x) + \beta y_2(x)$ and $\psi(x) = \gamma y_1(x) + \delta y_2(x)$ are linearly independent if and only if $\alpha\delta \neq \beta\gamma$.

Solution

We compute the Wronskian of the linear combinations:

$$W(\phi, \psi) = \begin{vmatrix} \phi & \psi \\ \phi' & \psi' \end{vmatrix} = \phi\psi' - \phi'\psi.$$

Substitute the definitions:

$$W(\phi, \psi) = (\alpha y_1 + \beta y_2)(\gamma y_1' + \delta y_2') - (\alpha y_1' + \beta y_2')(\gamma y_1 + \delta y_2).$$

Expand the terms:

$$(\alpha y_1 + \beta y_2)(\gamma y_1' + \delta y_2') = \alpha\gamma y_1 y_1' + \alpha\delta y_1 y_2' + \beta\gamma y_2 y_1' + \beta\delta y_2 y_2'.$$
$$(\alpha y_1' + \beta y_2')(\gamma y_1 + \delta y_2) = \alpha\gamma y_1' y_1 + \alpha\delta y_1' y_2 + \beta\gamma y_2' y_1 + \beta\delta y_2' y_2.$$

Subtracting these, the terms $\alpha\gamma y_1 y_1'$ and $\beta\delta y_2 y_2'$ cancel out:

$$\begin{aligned} W(\phi,\psi) &= \alpha\delta y_1 y_2' + \beta\gamma y_2 y_1' - \alpha\delta y_1' y_2 - \beta\gamma y_2' y_1 \\ &= \alpha\delta(y_1 y_2' - y_1' y_2) - \beta\gamma(y_1 y_2' - y_2 y_1') \\ &= (\alpha\delta - \beta\gamma)(y_1 y_2' - y_1' y_2). \end{aligned}$$

Recognizing $W(y_1, y_2) = y_1 y_2' - y_1' y_2$:

$$W(\phi,\psi) = (\alpha\delta - \beta\gamma)W(y_1, y_2).$$

Since y_1, y_2 are linearly independent solutions, $W(y_1, y_2) \neq 0$ everywhere. Therefore, $W(\phi,\psi) \neq 0$ (which means ϕ, ψ are independent) if and only if the determinant $\alpha\delta - \beta\gamma \neq 0$.

Problem 16

Show that any nontrivial solution $u(x)$ of $u'' + q(x)u = 0$, $q(x) < 0 \forall x$, has at most one zero.

Proof

We assume, for the sake of contradiction, that $u(x)$ has at least two distinct zeros, say at $x = a$ and $x = b$ with $a < b$, i.e., $u(a) = 0$ and $u(b) = 0$. Consider the comparison differential equation:

$$v'' + 0 \cdot v = 0.$$

A general solution to this equation is $v(x) = c_1 x + c_2$. We choose the specific solution $v(x) = 1$, which is strictly positive and has no zeros. We apply the **Sturm Comparison Theorem**. Let the coefficient of the first equation be $Q_1(x) = q(x)$ and the coefficient of the second equation be $Q_2(x) = 0$. The theorem states that if $Q_1(x) \leq Q_2(x)$, then between any two zeros of the solution to the first equation (u), there must be at least one zero of the solution to the second equation (v). Here, we are given $q(x) < 0$, so $Q_1(x) < Q_2(x)$ holds. Therefore, if $u(x)$ has zeros at a and b, the theorem guarantees that $v(x)$ must have a zero in the interval (a, b). However, $v(x) = 1$ is never zero. This is a contradiction. Thus, our initial assumption that $u(x)$ has two zeros must be false. A nontrivial solution can have at most one zero.

Problem 17

Let $u(x)$ be a nontrivial solution of $u'' + [1 + q(x)]u = 0$, $q(x) > 0$. Show that $u(x)$ has infinitely many zeros.

Proof

We compare the given ODE with the simple harmonic oscillator equation:

$$v'' + 1 \cdot v = 0.$$

Let $Q_1(x) = 1 + q(x)$ be the coefficient of the given ODE, and $Q_2(x) = 1$ be the coefficient of the comparison ODE. Since $q(x) > 0$, we have $Q_1(x) > Q_2(x)$ for all x. Consider a solution to the comparison equation, $v(x) = \sin x$. The zeros of $v(x)$ are at $x_n = n\pi$ for integers n. We apply the

Sturm Comparison Theorem. The theorem states that if $Q_1(x) \geq Q_2(x)$, then between any two consecutive zeros of the solution to the equation with the smaller coefficient (v), there must be at least one zero of the solution to the equation with the larger coefficient (u). The zeros of $v(x) = \sin x$ occur at $\ldots, -\pi, 0, \pi, 2\pi, \ldots$. Since there is an infinite sequence of such intervals $(n\pi, (n+1)\pi)$, and $u(x)$ must possess a zero within each such interval, it follows that $u(x)$ has infinitely many zeros.

Problem 18

Let $u(x)$ be a nontrivial solution of $u'' + q(x)u = 0$ on a closed interval $[a, b]$. Show that $u(x)$ has at most finitely many zeros in [a, b].

Proof

We proceed by contradiction. Assume that the nontrivial solution $u(x)$ has an infinite number of zeros in the closed interval $[a, b]$. Let $Z = \{x \in [a, b] : u(x) = 0\}$ be the set of these zeros. Since Z is an infinite subset of a bounded interval $[a, b]$, by the **Bolzano-Weierstrass Theorem**, the set Z must have an accumulation point (limit point) $x_0 \in [a, b]$. This implies there exists a sequence of distinct zeros $\{x_n\} \subset Z$ such that $\lim_{n\to\infty} x_n = x_0$. Since $u(x)$ is the solution to a differential equation with continuous coefficients, $u(x)$ is continuous. Therefore:

$$u(x_0) = \lim_{n\to\infty} u(x_n) = \lim_{n\to\infty} 0 = 0.$$

Thus, x_0 is also a zero. Furthermore, $u(x)$ is differentiable. We compute the derivative at x_0 using the definition of the derivative:

$$u'(x_0) = \lim_{x\to x_0} \frac{u(x) - u(x_0)}{x - x_0}.$$

Using the sequence $\{x_n\}$ to approach x_0:

$$u'(x_0) = \lim_{n\to\infty} \frac{u(x_n) - u(x_0)}{x_n - x_0} = \lim_{n\to\infty} \frac{0 - 0}{x_n - x_0} = 0.$$

We have established that at the point x_0, both $u(x_0) = 0$ and $u'(x_0) = 0$. The **Uniqueness Theorem** for linear initial value problems states that the only solution to $u'' + q(x)u = 0$ with $u(x_0) = 0$ and $u'(x_0) = 0$ is the trivial solution $u(x) \equiv 0$ for all x. This contradicts the hypothesis that $u(x)$ is a nontrivial solution. Therefore, the assumption that there are infinitely many zeros is false. The solution must have only finitely many zeros in $[a, b]$.